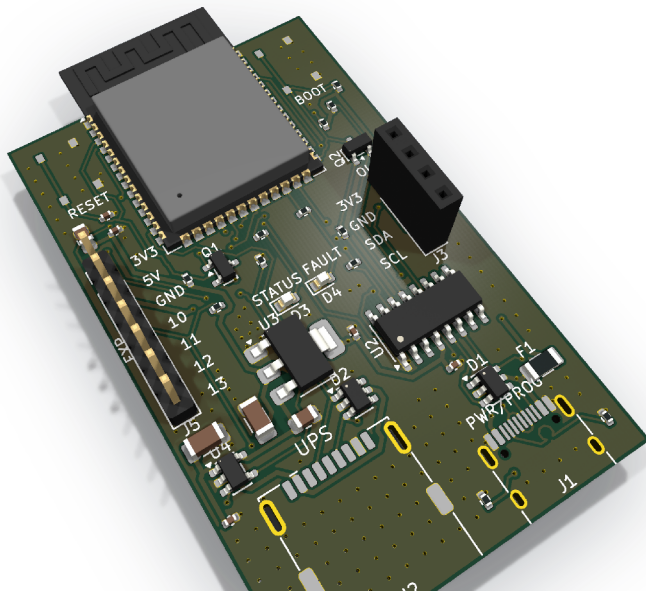


ups-adaptor

REV A · PRE-PRODUCTION

Your UPS on the network. No PC required.



A USB-only UPS is invisible to everything on your network. It needs a computer plugged into it, running all the time, just to say “the power is out”.

ups-adaptor is that computer, shrunk to 36 × 58 mm. It speaks **NUT on TCP 3493** — the protocol Home Assistant, Synology, TrueNAS, unraid and upsmon already understand — so nothing needs a plugin, a bridge or a driver.

Plug the UPS into the USB-A port, power the board from USB-C, and it appears on your network as **ups-adaptor.local**.

GENERIC BY DESIGN

Not built for one model

Parses the USB HID Power Device descriptor at runtime rather than hard-coding a UPS. Whatever your UPS reports, it forwards. Validated on a real APC Back-UPS RS 1000G: 86 fields decoded into 20 NUT variables, no vendor quirk needed.

RECOVERS ITSELF

Power-cycles a wedged UPS

VBUS to the UPS runs through a current-limited load switch on a GPIO. When a UPS interface locks up — and they do — firmware cycles its power instead of someone walking to the closet.

RECONNECTS

Survives the network

Wi-Fi drops are handled with a non-blocking retry, not a watchdog reboot. The board also survives the UPS being unplugged, replugged and power-cycled mid-session without a reset.

Technical specification

Controller	ESP32-S3-WROOM-1-N4 dual-core Xtensa LX7, 4 MB flash	Primary output	NUT server, TCP 3493 read-only protocol subset
Wireless	Wi-Fi 802.11 b/g/n, 2.4 GHz PCB antenna, Bluetooth disabled	Secondary output	MQTT with Home Assistant discovery
UPS port	USB-A host, USB 2.0 full speed 12 Mbps, HID Power Device class	Setup	Web UI over Wi-Fi; SoftAP for first-run provisioning
VBUS to UPS	Switched, 1 A current limit fault flag to firmware, 100 µF bulk	Discovery	mDNS — ups-adaptor.local
Power & console	USB-C: 5 V in, programming and serial on one connector	Expansion	4 × GPIO + 3V3 + 5V + GND I ² C header for a 128×64 OLED
Protection	ESD array on both USB ports 2 A resettable polyfuse	Board	36 × 58 mm, 2 layer surface mount, case-clip mounting